CSE340: Compter Architecture

Assignment 1

Total Points: 60

Deadline: March 1st, 2026 - 3:00 am.

- **Answer each question carefully and provide explanations wherever necessary.**
- **Submit your answers in a clear, organized format.**

Question 1 [10]

A semiconductor company manufactures two types of wafers:

- **Wafer A**: Diameter = 30 cm, cost = $30, defect density = 0.015 defects/cm², produces 150 dies.
- **Wafer B**: Diameter = 25 cm, cost = $25, defect density = 0.02 defects/cm², produces 120 dies.

Assume N = 2.

Tasks:

1. Explain the role of yield. [2]
2. Calculate the die area for each wafer (use total area / number of dies). [2]
3. Compute the yield for Wafer A and Wafer B. [2]
4. Determine the cost per functional die for both wafers. [2]
5. Which wafer is more cost-effective? Justify. [2]

Question 2 [8]

Two mobile processors (SnapDragon-X and MediaTech-Z) execute a program of 2 million instructions with the following breakdown:

Class	% of Instructions	CPI (SnapX)	CPI (MediaZ)
A	40%	2	1
B	30%	3	4
C	20%	1	2
D	10%	5	2

Clock rate:

- SnapX: 2.2 GHz
- MediaZ: 2.5 GHz

Tasks:

1. Compute the average CPI for both processors. [2]
2. Calculate execution time for each. [2]
3. Determine which processor has better performance and by what percentage. [2]
4. If the program takes 100 ms on a reference system, find the SPEC ratio for each processor. [2]

Question 3 [8]

Processor Q1 runs at 2.8 GHz with an average CPI of 1.8. You are asked to design a new processor (Q2) with the goal of improving performance by **40%** for the same ISA.

However, due to architectural changes, the average CPI increases to 2.2.

Tasks:

1. What execution time ratio does Q2 require to meet the performance goal? [3]
2. What clock rate must Q2 have to meet this goal? [2]

3. Is the trade-off between higher CPI and faster clock rate justified? Why or why not? [3]

Question 4 [4]

An existing processor operates with:

- Voltage: 1.2V, Frequency: 2.5 GHz, Capacitive load: 100 units

A newer version is introduced with:

- Voltage: 0.85V, Frequency: 2.2 GHz, Capacitive load: 80 units

Tasks:

1. Calculate the percentage reduction in power consumption. [2]
2. If the original system consumed 100W, how much power does the new system use? [2]

Question 5 [10]

Two embedded systems, **SysA** and **SysB**, run a benchmark.

- **SysA**: Executes 4 billion instructions, CPI = 1.5, Clock Rate = 3 GHz
- **SysB**: Executes 1.5 billion instructions, CPI = 2.0, Clock Rate = 2.5 GHz

Tasks:

1. Calculate execution time for both systems. [3]
2. Compute MIPS for both systems. [2]
3. Based on execution time, which system performs better? [2]
4. Explain why a higher MIPS does not always imply better performance.[3]

Question 6 [10]

Processor A executes:

- 25 instructions of Type-X (each takes 300 ps)
- 15 instructions of Type-Y (each takes 500 ps)

The reference system completes the same program in 14.2 ns.

Tasks:

1. Determine the total execution time for the processor. [2]
2. Calculate the individual CPI for each type of instruction and the average CPI assuming the clock period is 100 ps. [3]
3. Compute the SPEC ratio. [1]
4. To achieve a SPEC ratio of 1.5, what should be the new time for Type-X instructions? [4]

Question 7 [10]

1. A computer is running a program that requires 180s to be executed where 80s are required for R type instructions, 45s required for I/S type instructions and 55s required for SB type instructions. Can the performance be increased by 20% by only reducing the execution time for SB type instructions? If yes, determine the improvement factor. [3]
2. In the context of computing CPU performance, i) Instruction count, ii) CPI and iii) Clock rate are the main key factors that are considered. Which of these factors are affected by the choice of programming language and in what way? [3]
3. What do you understand by "Performance Via Prediction" in the context of computer architecture? Give a proper example of it. [2]
4. Why is it necessary to keep redundancy while designing a system? Explain with a scenario where redundancy is very helpful. [2]

Submission Guidelines

Submit your scanned PDF of handwritten assignment in the provided submission link.